NR2DT-06: Step 6 — Migrate Synthetics, SLOs & Workloads

> **Series:** NR2DT — New Relic to Dynatrace Migration Steps | **Notebook:** 6 of 10 | **Created:** April 2026 | **Last Updated:** 11/03/2026

Overview

Goal of this step: complete Wave 2 (synthetics) and Wave 4 (SLOs + workloads). Synthetics are external-facing and visible quickly; SLOs need a 7-day delta-validation window before the math is trusted.

Procedural — see **NRLC-05** (Synthetic Migration) and **NRLC-06** (SLO & Workload Migration) for component depth.

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Prerequisites

Requirement	Details
Audience	Migration lead + assigned engineer for this step
Completed	NR2DT-05 — Migrate Dashboards & Alerts

Requirement	Details
Format	Procedural step — use as a runbook; defer to NRLC for depth
NRLC deep dives	NRLC-05, NRLC-06

1. Wave 2 — Synthetics

Order: HTTP first (lowest risk), then Browser, then API multi-step. Scripted browsers are out of band — they get rebuilt manually.

```
# Preview: export + transform + diff — imports NOTHING (see NR2DT-05 Wave 1)
python3 migrate.py migrate --dry-run --diff --components synthetics --output
./run-synthetics

# Import: the same command without --dry-run
python3 migrate.py migrate --components synthetics --output ./run-synthetics
```

> **Check what the import actually created.** At tool v3.0.0 (commit `78cbfce`) the default (Gen3) import phase marks synthetic tests **SKIPPED** with the reason *Gen3 synthetic emission requires per-facet settings objects*; the transformed monitors are still written to `./run-synthetics/transformed/dynatrace_config.json`. Read the run's summary table and skip reasons, and create any skipped monitors from that file until your tool version imports them.
>
> **Sources:** migrate.py @ 78cbfce (tool repo, GitHub), read 10/05/2026.

Re-enter secrets in the DT credentials vault for any monitor that uses bearer tokens, basic auth, or OAuth.

Scripted browser monitors:

1. The transformer emits a stub Browser Monitor with the target URL
2. The original NR script is **not** copied: the stub carries a `migrated-from: newrelic` tag and a placeholder JavaScript step (`// TODO: Add custom validation from New Relic script`), and the run prints a warning naming the monitor's URL
3. An engineer rebuilds the user journey using DT's clickpath recorder or DSL
4. Validation: confirm the new clickpath catches the same regressions during a 1-week dual-run

> **Sources:** transformers/synthetic_transformer.py @ 78cbfce (tool repo, GitHub), read 11/03/2026.

W2 — Synthetics SLA: dual-run for ≥ 1 week. DT availability % matches NR within ±0.5%. If delta exceeds, investigate location coverage and credential validity first.

2. Wave 4 — SLOs & Workloads

SLOs depend on metric expressions being mathematically equivalent. Workloads become OpenPipeline enrichments + IAM bucket conditions (Gen3 pattern).

```
# Preview (imports nothing), then import only SLOs + workloads from the
previewed file
python3 migrate.py migrate --dry-run --diff --components slos,workloads --
output ./run-slos-workloads
python3 migrate.py migrate --import-only --input ./run-slos-workloads --
components slos,workloads
```

`slos` depends on `alerts`, and a full `migrate` run adds dependencies automatically — `--components slos,workloads` without `--dry-run` would export, transform and import alerts and notification channels again. `--import-only` imports only the components named, but writes **no** rollback manifest; record what it created from its summary. At tool v3.0.0 the default (Gen3) import marks workload segments and IAM policies **SKIPPED** (the Platform segment API and the Account Management IAM API are not wired into the tool), so plan to create workload scoping yourself per NR2DT-03.

Run the SLO auditor — a metric-validity check, not an SLI comparison:

```
export DYNATRACE_OAUTH_TOKEN="&lt;OAuth bearer token&gt;"   # audit-slos exits
without it
python3 migrate.py audit-slos
```

`audit-slos` reads every SLO in the tenant through the Platform SLO API and reports `OK` / `FAIL` with issues per SLO (missing metrics, invalid aggregations, NRQL that was not converted), then a `valid / total` count. It needs `DYNATRACE_ENVIRONMENT_URL` and `DYNATRACE_OAUTH_TOKEN`. It computes no NR or DT SLI and takes no window or tolerance.

The 7-day SLI delta is computed by hand. For each SLO, take NR's SLI for `SINCE 7 days ago` and the DT SLO's evaluated value over the same 7 days, and record the delta. Any SLO with delta > 0.5% needs investigation:

1. Check unit conversion (ms vs. s vs. ns)
2. Check entity scope — does the bucket / OpenPipeline enrichment filter capture the same entities as NR's workload?
3. Check time alignment — NR's `SINCE 7 days ago` may bin differently than DT's `from:-7d`

W4 — SLO delta: all SLOs within tolerance for ≥ 7 days (measured as above), and `audit-slos` reports every SLO valid.

> **Sources:** migrate.py @ 78cbfce (tool repo, GitHub), config/settings.py @ 78cbfce (tool repo, GitHub), read 10/05/2026.

3. Step Exit Criteria

S6 — Synthetics + SLOs Migrated

- [ ] All synthetics migrated; W2 dual-run availability ±0.5%
- [ ] Scripted browsers rebuilt and validated
- [ ] All synthetic credentials re-entered in DT vault
- [ ] All SLOs migrated; W4 7-day delta ≤ 0.5%; `audit-slos` clean
- [ ] Workload identifiers visible as OpenPipeline-enriched attributes in DQL queries

Next step: NR2DT-07 — Migrate Logs, Tags & Drop Rules.

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